

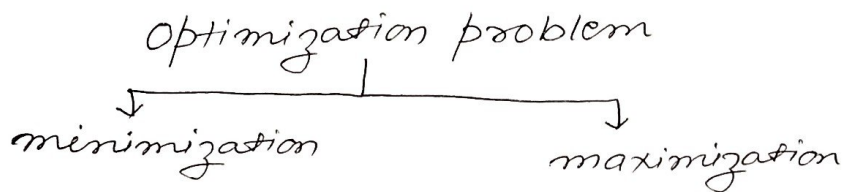
Ch-1 INTRODUCTION TO OPTIMIZATION

→ "Optimization is the act of obtaining the best result under given circumstances."

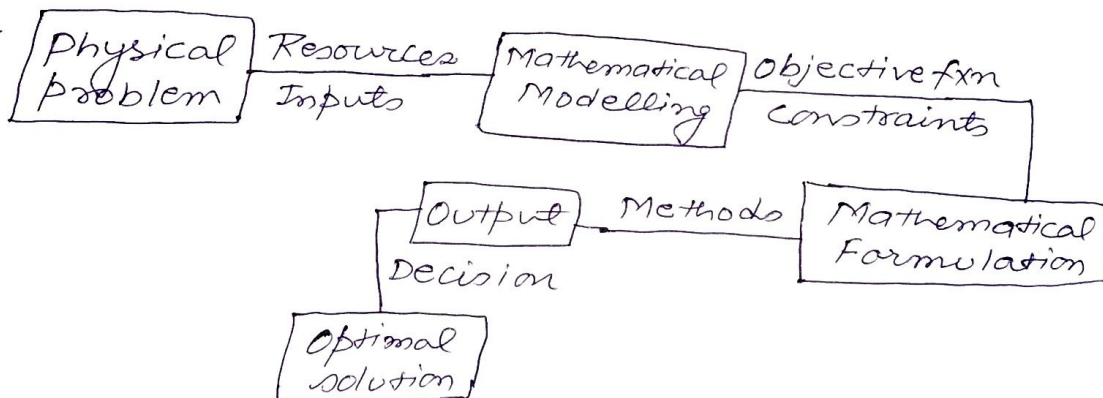
→ "Optimization can be defined as the process of finding the conditions that give the maximum or minimum value of a fcn."

Operational Research → is a branch of mathematics concerned with the application of scientific methods and techniques to decision making problems with establishing the best or optimal solutions.

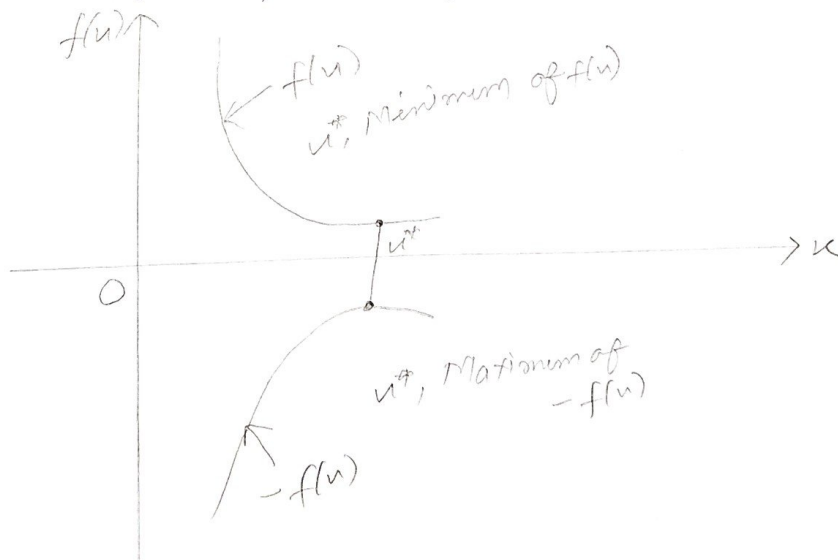
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→ If $f(x)$ is given; set of x values - outputs.



→ If a point u^* corresponds to the minimum value of $f(x)$, $f(u)$, the same point also corresponds to the maximum value of the negative of the $f(x)$, $-f(u)$.

Engineering Applications of Optimization:-

- (1) Design of aircraft and aerospace structures for minimum weight.
- (2) Design of civil engineering structures such as frames, foundations, bridges, towers, chimneys, and dams for minimum cost.
- (3) Minimum-weight design of structures for earthquake, wind, and other types of random loading.
- (4) Design of water resources systems for maximum benefit.
- (5) Optimal plastic design of structures.
- (6) Optimal design of electrical networks.
- (7) Shortest route taken by a salesperson visiting various cities during one tour.
- (8) Optimum design of chemical processing equipment and plants.
- (9) Selection of a site for an industry.
- (10) Planning of maintenance and replacement of equipment to reduce operating costs.

Q.1) Same Q. as of Mam.

Ans) Let,
 u_1 be the no. of M_1 models
 u_2 be the no. of M_2 models

The profit per week
 $Z = 3u_1 + 4u_2$

Per week the grinding hours
 $4u_1 + 2u_2$

Per week polishing hours
 $2u_1 + 5u_2$

Maximum no. of hours of grinding, $2 \times 40 = 80$
 — polishing, $3 \times 60 = 180$

Maximize $Z = 3u_1 + 4u_2$
 subject to $4u_1 + 2u_2 \leq 80$
 $2u_1 + 5u_2 \leq 180$, $u_1, u_2 \geq 0$

Solⁿ by graphical method:—

$L_1: 4u_1 + 2u_2 = 80$

$L_2: 2u_1 + 5u_2 = 180$

u_1	u_2
0	40
20	0

for L_1

u_1	u_2
0	36
90	0

for L_2

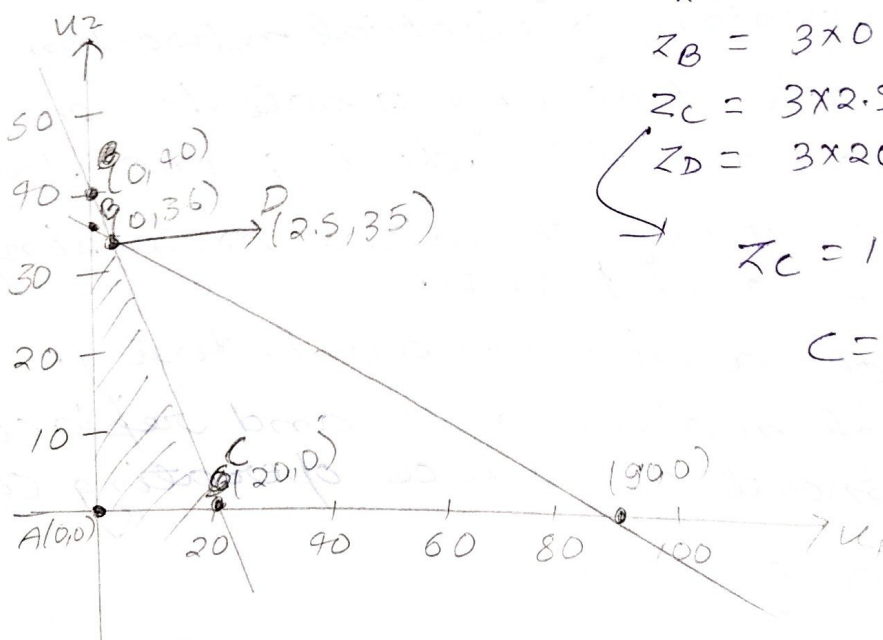
Now, $Z_A = 3 \times 0 + 4 \times 0 = 0$

$Z_B = 3 \times 0 + 4 \times 36 = 144$

$Z_C = 3 \times 2.5 + 4 \times 35 = 147.5$

$Z_D = 3 \times 20 + 4 \times 0 = 60$

$Z_C = 147.5$
 $C = (2.5, 35)$



$A(0,0)$, $B(0,36)$
 $C(2.5,35)$, $D(20,0)$

Single Variable Optimization:—
 next lecture.

LINEAR PROGRAMMING PROBLEM:-

→ "It is the method used in decision making in business for obtaining the maximum or minimum value of linear expression subject to satisfy certain given linear inequations."

Objective function → A linear function of the involve variable which we want to maximize or minimize subject to given linear constraints is known as Objective function. It is represented by Z .

It is of the form

$$Z = ax + by$$

Linear Constraints → The limitations or restrictions on the resources can often be expressed in the form of linear inequations known as linear constraints.

Optimal Value of an Objective function:-

The max. & min. value of an obj. fxn is known as ~~of~~ its optimal value.

Feasible Solution → The set of values of variable satisfy all constraints is known as feasible solution of system of inequation.

Optimal solution → A feasible solution which leads to optimal value of an objective fxn is known as optimal solⁿ of system of inequation.

Kinds of Region:

(a) Bounded Region

(1) Obj. fxn (2) Constraints (3) plot the soln region

(4) Corner points is the answer.

(b) Unbounded Region

(1) Obj. fxn (2) Constraints (3) plot the soln region

(4) Corner points is the answer

(5) Verification

Q1: Solve the LPP graphically:

Maximize $Z = 4x + y \rightarrow$ obj. fxn

Subject to Constraints: $x \geq 0, y \geq 0$
 $x + y \leq 50$

Ineqvation	Equality	Double intercept form	Shaded region
$x + y \leq 50$ $x \geq 0$ $y \geq 0$	$x + y = 50$	$\left(\frac{x}{a} + \frac{y}{b} = 1 \right)$ $\rightarrow \frac{x}{50} + \frac{y}{50} = 1$	$0 + 0 \leq 50$ $0 \leq 50$ <u>origin satisfy</u> <u>origin side</u>

$\rightarrow$ if $(0,0)$ satisfy $\rightarrow$ origin side
 if $(0,0)$ not satisfy $\rightarrow$ Non-origin side

Now, plot on graph:-

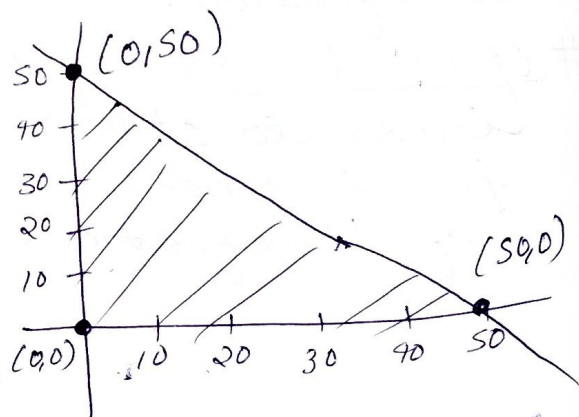
Now, checking each points by putting in Z

$(0,0) \rightarrow Z_1 = 0$

$(50,0) \rightarrow Z_2 = 200 \rightarrow$ maximum

$(0,50) \rightarrow Z_3 = 50$

Max Z is at $x=50, y=0$



Q2) Solve the LPP graphically:

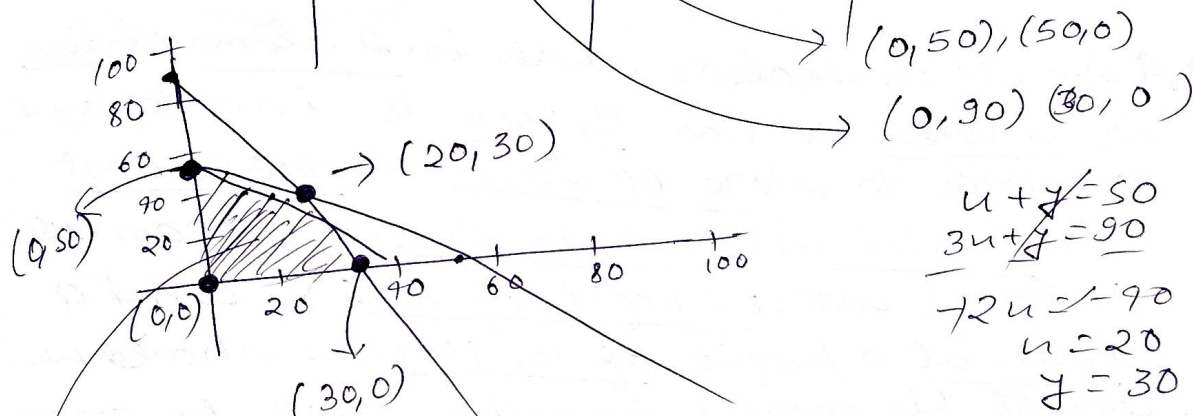
Maximize $Z = 4u + 7$

STC: $u \geq 0, y \geq 0$

$u + y \leq 50$

$3u + y \leq 90$

Ar) Inequality	Equality	D.I. form	Shaded region
$u + y \leq 50$	$u + y = 50$	$\frac{u}{50} + \frac{y}{50} = 1$	$0 + 0 \leq 50$
$3u + y \leq 90$	$3u + y = 90$	$\frac{u}{30} + \frac{y}{90} = 1$	$0 \leq 90$



$(0,0) \rightarrow Z_1 = 0$
 $(30,0) \rightarrow Z_2 = 120$
 $(0,50) \rightarrow Z_3 = 50$
 $(20,30) \rightarrow Z_4 = 110$

Maximum at $u = 30$
 $y = 0$.

- Shaded region is called feasible region.
- Points inside feasible region feasible solⁿ.
- Points outside feasible region infeasible solⁿ.
- Points in the feasible region that gives the optimal value of Z : Optimal (feasible) solⁿ.

Formulation of LPP:-

Step 1: Identify the unknowns in the given LPP.
Denote them by u & y .

Step 2: Formulate the obj. fcn in terms of u & y , which is to be maximized or minimized.

Step 3: Translate all constraints in the form of linear in equations.

Step 4: Solve these inequations simultaneously.
Mark the common area by shaded region.
This is the feasible region.

Step 5: Find the coordinate of all the vertices of the feasible solution.

Step 6: Find the value of obj. fxn at each vertex of feasible region.

Step 7: Find the values of u & y for which the obj. fxn $Z = au + by$ has min. or max value.

Q3) A furniture dealer deals in 2 items chairs or tables. He has Rs 5000 to invest and a space to store 60 items. A table cost him Rs 250 and Chair Rs 50. He can sell a table at a profit of Rs 50, and a chair at a profit of Rs 15. How should he invest his money in order that he may maximize the profit. Formulate LPP & solve graphically.

Ans)

items	Cost	Profit
table	250	50
Chair	50	15

Let u tables &
 y chairs purchase

$$\text{Max} = 50u + 15y$$

$$\text{STC: } 250u + 50y \leq 5000$$

$$u + y \leq 60$$

Now, solve.

Q4) A merchant plans to sell two types of PCs. A desktop model and a portable model that will cost Rs 25000 and Rs 40000 respectively. He estimated that the total monthly demand of Computers will not exceed 250 units. Determine the number of units of each type of Computers which

the merchant should stock to get maximum profit if he does not want to invest more than Rs 70 lakhs and its profit on desktop model is Rs 4500 and on portable model is 5000. Make an LP & solve graphically.

Ans)

Type	Cost	Profit
Desktop	25000	4500
Laptop	40000	5000

Profit fn, maximize $Z = 4500u + 5000j$

Max required units = 250 units

Max amt. = 7000000

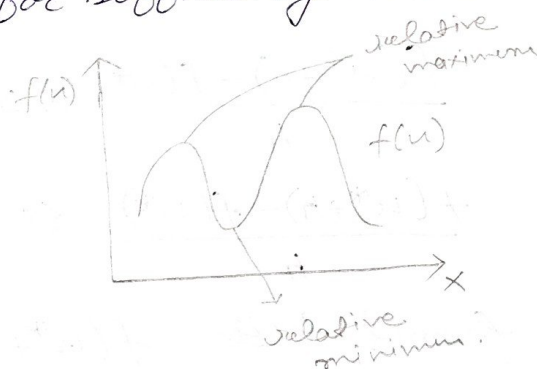
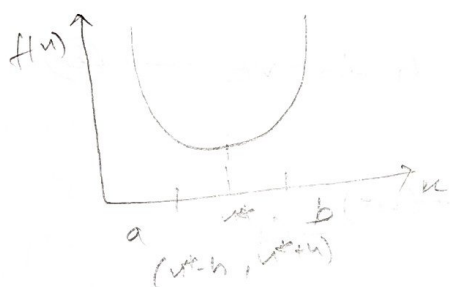
$$u + j \leq 250$$

Let u be desktop model & j portable model sold.

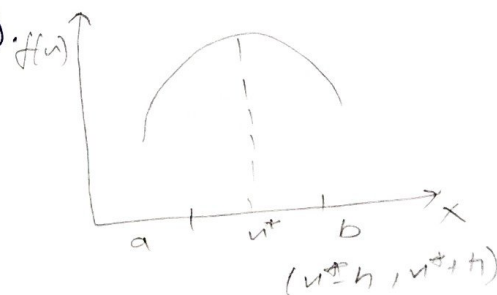
$$25000u + 40000j \leq 7000000$$

Single Variable Optimization:-

- (1) Let $f(u)$ be a fn defined on $[a, b]$. Then $u = u^*$ is said to be relative minimum of $f(u)$ if
- $$f(u^*) \leq f(u^* + h)$$
- for sufficiently small +ve or -ve value of h .



- (2) $u = u^*$ is said to be relative maximum if
- $$f(u^*) \geq f(u^* + h)$$
- for sufficiently small values of h (+ve or -ve value).



(3) $u = u^*$ is said to be global minimum if $f(u) \geq f(u^*)$ for all u in the domain.

(4) $u = u^*$ is said to be global maximum if $f(u) \leq f(u^*)$ for all u in the domain.

Necessary Condition

If $f(u)$ be a single variable fn over $[a, b]$ & $u = u^*$ is said to be relative minimum for $a < u^* < b$, $\frac{df(u)}{du} = f'(u)$ exist with finite number then $f'(u) = 0$.

Proof:-

$$f'(u^*) = \lim_{h \rightarrow 0} \frac{f(u^*+h) - f(u^*)}{h} \text{ exists.}$$

$$\text{Claim: } f'(u^*) = 0$$

$u = u^*$ be the relative minimum point.

$$f(u^*) \leq f(u^*+h)$$

$$\frac{f(u^*+h) - f(u^*)}{h} \geq 0 \text{ if } h \text{ is +ve} \text{ --- (i)}$$

$$\frac{f(u^*+h) - f(u^*)}{h} \leq 0 \text{ if } h \text{ is -ve} \text{ --- (ii)}$$

$$\text{from (i) \& (ii) } \frac{f(u^*+h) - f(u^*)}{h} = 0$$

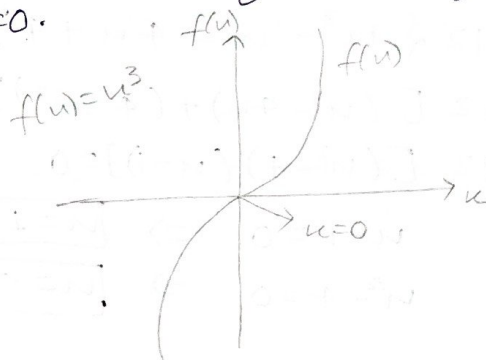
$$f'(u^*) = 0 \text{ proved}$$

$f(u)$ on $[a, b]$

$u = u^*$ is the relative minimum $a < u^* < b$

- (1) We cannot say the nature of u^* at the end points. $f'(u^*) = 0$
- (2) If $f'(u)$ is undefined or doesn't exist at $u = u^*$, then we cannot conclude the relative minimum & relative maximum of $f(u^*)$ at u^* .
- (3) We cannot conclude the relative minimum and relative maximum of $u = u^*$ for some $f(x)$ even $f'(u^*) = 0$.

$$\begin{aligned} f(u) &= u^3 \\ f'(u) &= 3u^2, \\ 3u^2 &= 0 \\ \Rightarrow u &= 0 \end{aligned}$$



$\rightarrow u=0$ is neither relative min. nor relative max. or it is called point of inflection.

Sufficient Condition for single Variable $f(x)$.

If $f'(u^*) = f''(u^*) = \dots = f^{(n-1)}(u^*) = 0$ and $f^{(n)}(u^*) \neq 0$ then $u = u^*$ is said to be

- (i) relative minimum if $f^{(n)}(u^*) > 0$ & n is even.
- (ii) relative maximum if $f^{(n)}(u^*) < 0$ & n is even.
- (iii) point inflection if n is odd.

Proof:-
$$f(u^* + h) = f(u^*) + hf'(u^*) + \frac{h^2}{2}f''(u^*) + \dots + \frac{h^n}{n!}f^{(n)}(u^*)$$

$$f(u^* + h) = f(u^*) + \frac{h^n}{n!} f^{(n)}(u^*)$$

$$\Rightarrow f(u^* + h) - f(u^*) = \frac{h^n}{n!} f^{(n)}(u^*)$$

Q. Find the optimal points and the optimality of the following fn.

(i) $f(u) = 3u^4 - 4u^3 - 24u^2 + 48u + 15$

(ii) $f(u) = 12u^5 - 45u^4 + 40u^3 + 5$

Ans) (i) Given,

$$f(u) = 3u^4 - 4u^3 - 24u^2 + 48u + 15$$

$$f'(u) = 0$$

$$\Rightarrow 12u^3 - 12u^2 - 48u + 48 = 0 \quad \checkmark$$

$$\Rightarrow 12 \{ u^3 - u^2 - 4u + 4 \} = 0$$

$$\Rightarrow 12 [(u^3 - 4u) + (4 - u^2)] = 0$$

$$\Rightarrow 12 [(u^2 - 4)(u - 1)] = 0$$

$$u - 1 = 0 \Rightarrow \boxed{u = 1}$$

$$u^2 - 4 = 0 \Rightarrow \boxed{u = -2 \text{ or } 2}$$

$$f''(u) = 36u^2 - 24u - 48 \quad \text{--- (i)}$$

At $u = 1$ in (i)

$$f''(u) = 36 \times 1^2 - 24 \times 1 - 48 \\ = -36 < 0$$

$u = 1$ relative maximum

At $u = -2$ in (i)

$$f''(u) = 36 \times (-2)^2 - 24 \times 2 - 48 \\ = 36 \times 4 + 98 - 98 \\ = 144 > 0$$

$u = -2$ is the relative minimum.

At $u = 2$ in (i)

$$f''(u) = 36 \times (2)^2 - 24 \times 2 - 48 \\ = 144 - 98 - 98 \\ = 98 > 0$$

$u = 2$ is also a relative minimum.

(ii) Given,

$$f(u) = 12u^5 - 45u^4 + 40u^3 + 5$$

$$f'(u) = 0$$

$$\Rightarrow 60u^4 - 180u^3 + 120u^2 \quad \checkmark$$

$$\Rightarrow 60u^2 [u^2 - 3u + 2] = 0$$

$$\therefore u^2 = 0 \Rightarrow \boxed{u=0}$$

$$u^2 - 3u + 2 = 0$$

$$\Rightarrow \boxed{u=1 \mid u=2}$$

At $u=1$

$$f''(u) = 240u^3 - 540u^2 + 240u \quad \text{--- (i)}$$

At $u=0$ in (i)

$$\begin{aligned} f''(0) &= 240 \times 0 - 540 \times 0 + 240 \times 0 \\ &= \underline{\underline{0}} \end{aligned}$$

At $u=1$ in (i)

$$\begin{aligned} f''(1) &= 240 \times (1)^3 - 540 \times (1)^2 + 240 \times 1 \\ &= 240 - 540 + 240 \\ &= 480 - 540 = -60 < 0 \end{aligned}$$

$u=1$ is relative maximum.

At $u=2$ in (i)

$$\begin{aligned} f''(2) &= 240 \times (2)^3 - 540 \times (2)^2 + 240 \times 2 \\ &= 240 \times 8 - 540 \times 4 \\ &= 2400 - 2160 \\ &= 340 > 0 \end{aligned}$$

$u=2$ is the relative minimum.

$$\therefore f'''(u) = 3 \times 240u^2 - 2 \times 540u + 240$$

$$f'''(0) = 240$$

if the odd derivative is non-zero. So,
at $u=0$ will be pt. of inflection.